THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 11: **Keeping Time with the Skies** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A vertical stick of 1 m casts a shadow of 1 m long at some time. The Sun's elevation at that time is about: 1

(a) 60 degrees **(b)** 45 degrees **(c)** 30 degrees **(d)** 90 degrees

2. A lunar year consists of 12 lunar months of 29.5 days each. The number of days is: 1

(a) 365 **(b)** 354 **(c)** 348 **(d)** 360

3. The Moon's waxing crescent is seen in the: 1

(a) north at dawn **(b)** east at midnight
(c) south at noon **(d)** west after sunset

4. At noon the shadow of a stick is directly towards the north (in India). The Sun then is: 1

(a) to the south of the observer **(b)** overhead
(c) to the north **(d)** below the horizon

5. The difference between the solar year (365 days) and the lunar year (354 days) is about: 1

(a) 1 day **(b)** 11 days **(c)** 5 days **(d)** 30 days

6. The full Moon rises at about: 1

(a) sunrise **(b)** midnight **(c)** sunset **(d)** noon

7. If today is a full moon, the next new moon will be after about: 1

(a) 30 days **(b)** 15 days **(c)** 7 days **(d)** 60 days

8. After about 3 years, the lunar calendar falls behind the solar by about: 1

(a) 11 days **(b)** 22 days **(c)** 33 days **(d)** 1 day

9. The new Moon rises at about: 1

(a) sunset **(b)** noon **(c)** sunrise **(d)** midnight

10. If the new moon is on 1 January, the next full moon will be on about: 1

(a) 15 January **(b)** 1 February **(c)** 8 January **(d)** 30 January

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A stick of height 2 m casts a shadow of 2 m at 10 am. What is the angle of the Sun above the horizon? 2

12. Why does the lunar year have 354 days whereas the solar year has 365 days? 2

13. If the shadow of a tree is 6 m when a 2 m stick has a shadow of 1 m, find the height of the tree. 2

14. Why is the Pole Star not visible from the southern hemisphere? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student marks the tip of the shadow of a 1 m stick at 9 am, 12 noon and 3 pm. Describe the pattern and the shadow lengths relationship. 3

16. Compare the Gregorian, Islamic and Hindu calendars on basis of the Sun, Moon and year length. 3

17. Why is the day longer in summer? Explain with the tilt of the Earth. 3

OR

Explain how the shadow of a stick can be used to find the north-south direction at noon. 3

18. Describe how the position of the Pole Star helps a traveller find latitude. 3

OR

What is the importance of the Moon in Indian festivals and in tides? Give two examples. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe an activity to find the time of local noon using the shadow of a stick and explain how you find north-south. 5

OR

A traveller finds the Pole Star 25 degrees above the northern horizon. Find his latitude and describe how one can locate the Pole Star with the Big Dipper. 5

20. Describe the lunar phases through a month with the position of the Moon relative to the Sun and Earth, and the times of rising and setting. 5

OR

Explain why the length of the shadow at noon changes through the year and relate it to the seasons. 5

END OF QUESTION PAPER — Total: 40 Marks

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